Assignment 7: Microprogrammed Control Unit Design

PCC CS-401: Computer Organization and Architecture

August 7, 2026

Answer every question. Show all working for Numerical and Design questions.

7.1 Conceptual

Q1. Explain the role of the mapping unit in a microprogrammed control unit. What would happen if the mapping unit were removed and MPC simply reset to address 0 after every END?

Q2. Compare horizontal and vertical microinstruction encoding in terms of (a) control-word width and (b) the delay before control signals become valid after CM is read.

7.2 Numerical

Q3. A machine has 20 mutually-exclusive bus-source signals (exactly one is active at a time). How many bits are needed to encode this as a single vertical field? How many bits would the same 20 signals need if encoded horizontally (one bit per signal)?

Q4. ADD requires 3 control-memory rows (as in the lecture's worked example) and SUB requires a similarly-shaped 3-row microprogram. A third instruction, a register-only NOT R1 (one-operand, no memory or bus-buffering needed beyond reading and writing R1), requires 2 rows. Assuming no rows are shared between instructions, how many total control-memory rows are needed to hold all three microprograms?

7.3 Design

Q5. Write the microprogram (microaddress, RTL, control word, next-address field) for R4 $\leftarrow$ R5 $-$ R6 on the single-bus datapath, in the same table format as the lecture's worked example for R1 $\leftarrow$ R2+R3. Your microprogram must have 3 rows, correct next-address fields, and an END marker on the last row.

Q6. The worked ADD microprogram in the lecture uses 8 distinct control signals: $R2_{\text{out}}, R3_{\text{out}}, Y_{\text{in}}$, ALU=add, Z_{in}, Z_{out}, $R1_{\text{in}}$, END. Design a vertical encoding scheme for these 8 signals by grouping mutually-exclusive signals into encoded fields (following the lecture's worked example, which grouped bus-source, destination, and ALU-operation into three fields). State your field groupings and the resulting total control-word width in bits, compared to the 8 bits horizontal encoding would require.